

Find a Problem

TODAY YOU'LL MAKE

A list of problems worth fixing.

GET READY

- ☐ This book
- ☐ Pencil
- ☐ Two people to ask

STEPS

- 1** Walk into every room in the house with this book.
- 2** Write down **10 things** that are annoying, slow, or easy to forget.
- 3** Ask two people what annoys them. Add theirs to the list.
- 4** Cross out any you can't build with what's on the shelf.
- 5** Circle your favourite **three**.

IF IT BREAKS

Can't think of anything? Watch someone do a boring job. Wait for the sigh.

Ideas too big? Shrink them. Not a robot — a light that says the dog was fed.

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YOUR CHALLENGE

Pick the one that helps someone else, not you. Those are the best inventions.

YOUR 10 PROBLEMS**I DID IT!**

Date: _____

ONE THING I LEARNED

Sketch the Fix

TODAY YOU'LL MAKE

Three drawings, and one winner.

GET READY

- ☐ This book
- ☐ Pencil
- ☐ Someone to show

STEPS

- 1 Draw the problem at the top of the page.
- 2 Draw **three different** machines that fix it.
- 3 Label the parts of each: sensor, light, buzzer, motor.
- 4 Show all three to someone. Ask which they'd use.
- 5 Circle the winner. List every part you need.

IF IT BREAKS

Only one idea? Draw the lazy version, the expensive version, and the tiny version.

Can't draw? Boxes and arrows count. Nobody is marking the picture.

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YOUR CHALLENGE

Draw the silliest one too. Sometimes that's the one that works.

YOUR THREE IDEAS**I DID IT!**

Date: _____

ONE THING I LEARNED

Build the Ugly Version

TODAY YOU'LL MAKE

A rough build that actually works.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ Your parts list
- ☐ Cardboard + tape
- ☐ Pencil

STEPS

- 1 Lay every part from your list out on the table.
- 2 Wire the sensor first. Get its number onto the screen.
- 3 Add the thing that happens — light, sound, or servo.
- 4 Hold it together with cardboard and tape. Don't make it pretty.
- 5 Make it work **once**, start to finish.

IF IT BREAKS

Everything broken at once? Unplug all but the sensor. Add one part back at a time.

Worked, now it doesn't? A wire has moved. It is almost always a wire.

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YOUR CHALLENGE

Time yourself. Can you get from nothing to working in an hour? Ugly is allowed.

DRAW HOW IT GOES TOGETHER

**I DID IT!**

Date: _____

ONE THING I LEARNED

Test It on Someone

TODAY YOU'LL MAKE

One real fix, found by watching someone else.

GET READY

- ☐ Your invention
- ☐ Two testers
- ☐ Pencil

STEPS

- 1 Hand it to someone. Say **nothing**.
- 2 Write down every place they get stuck.
- 3 Ask them: what did you think it would do?
- 4 Fix the **one** worst thing on your list.
- 5 Hand it to a second person. See if that thing is gone.

IF IT BREAKS

They didn't know what to do? That is the result. Add a label, a light, or an instruction.

It broke during the test? Also a result. Tape the weak part and test it again.

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YOUR CHALLENGE

Test it on someone younger than you. They say what adults are too polite to say.

WHERE THEY GOT STUCK**I DID IT!**

Date: _____

ONE THING I LEARNED

Invention Night

TODAY YOU'LL MAKE

Your invention, shown to a room full of people.

GET READY

- ☐ Your invention
- ☐ Spare batteries
- ☐ Card for a sign
- ☐ This book

STEPS

- 1 Make a sign: the problem, your name, what it does.
- 2 Say your demo out loud **three times** on your own first.
- 3 Set it up on a table with spare batteries ready.
- 4 Invite everyone. Show it. Then let them try it.
- 5 Flip back through this book. All 25 weeks.

IF IT BREAKS

Won't work in front of people? Fresh batteries, and a photo of it working as back-up.

Nervous? Say the problem first, then show it. Thirty seconds is plenty.

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YOUR CHALLENGE

Teach one person one thing you learned, well enough that they can do it without you.

DESIGN YOUR SIGN**I DID IT!**

Date: _____

ONE THING I LEARNED
